

FINAL REPORT

Quantifying human drivers of noise in the Channel Islands National Marine Sanctuary: a focus on seal bombs

Investigators and contact details

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This report follows a consistent formatting structure. For each section, regular text appears first as a reminder of what was proposed, and it is followed by bolded text, tables and figures that serve as activity summaries and description of results. So in general bolded text is new, and regular text is not. The project duration was Jan. 2018-Dec. 2019 (Fig. 1).

OAP Project: Seal Bombs in CINMS

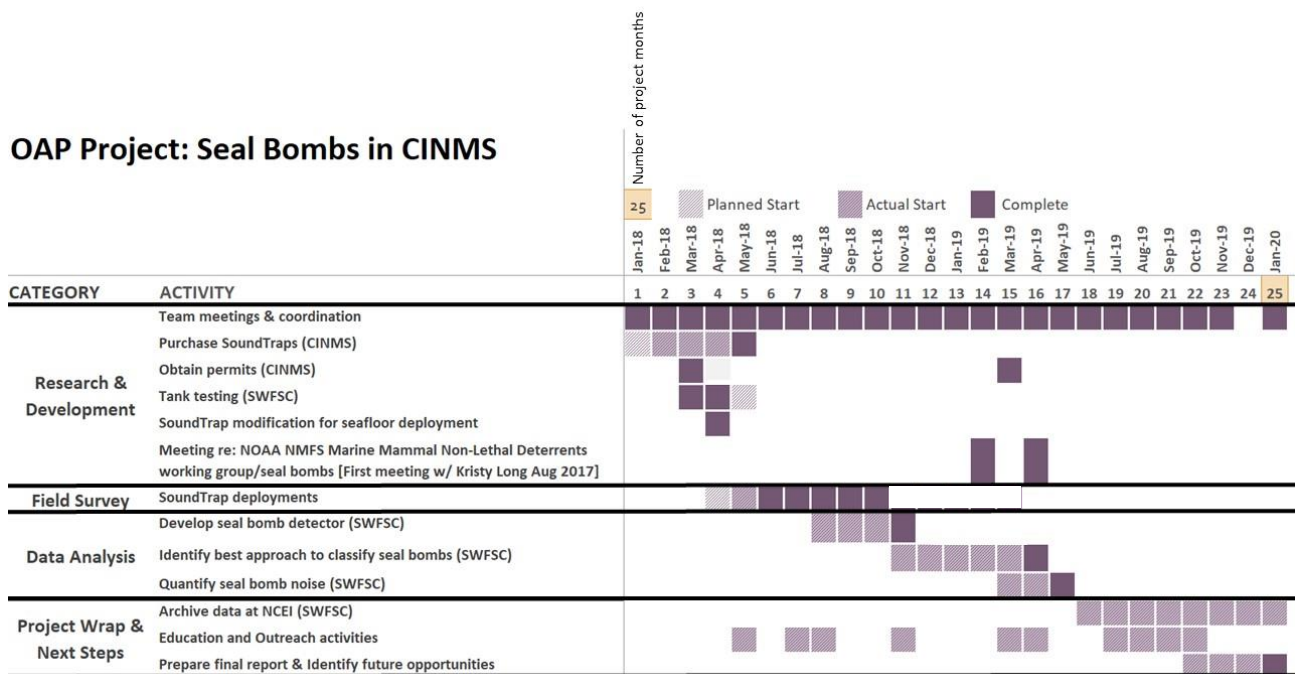


Figure 1. Color-coded project management chart showing the timing of activities described in more detail in the report that follows.

Data Collection

1.1 Obtain permits (SWFSC & CINMS)

CINMS research permits to discharge equipment in sanctuary waters and to disturb the seafloor will be acquired. Issued permits may include conditions not outlined in this proposal.

Authorized to operate under the Superintendent's Permits #CINMS-2014-001 and #CINMS-2019-001.

1.2 Initial testing of SoundTrap at SWFSC Technology Tank (SWFSC)

The SoundTrap 300HF is an autonomous single channel recorder with a 576 kHz sampling rate. Recording duration varies based on sampling rate, duty cycle vs. continuous recording, and data and power storage limitations. We estimate that we will be able to record for a minimum of 5 weeks. Hardware testing will ensure maximum recording duration without sacrificing stability. Testing will be conducted at the SWFSC Ocean Technology Development Tank in La Jolla, CA. A duty cycle may be considered to extend recording duration.

We were able to purchase two high frequency SoundTrap300s (ST300), one of which is a high frequency recorder, by joining a NOAA procurement facilitated by Jennifer Keating from NMFS PIFSC. Preliminary duty cycle estimates were based on the manufacturer's endurance calculator for the ST300HF with external battery pack (sample rate 96kHz with duty cycle 30 min on every 60 minutes). Extended testing was conducted off San Diego, which determined that the number of click detections (snapping shrimp, primarily) may decrease the overall endurance time due to increased memory usage for click files. Also, the compression for low-noise areas is more efficient than for high-noise areas, so the endurance time may be increased. As the ambient noise and presence of snapping shrimp vary by location, we were unable to do more than estimate the overall endurance time.

1.3 Modify SoundTrap for seafloor deployment (SWFSC)

SoundTraps will be tethered to a seafloor mooring and retrieved using acoustic releases. We will follow recommendations outlined in Dudzinski et al. (2011), as well as advice from experienced colleagues (Marc Lammers, Steve Dawson). We have secured two acoustic releases for loan from Brandon Southall and Karin Forney. We plan to deploy the secondary acoustic releases on both SoundTraps for redundancy and retrieval assurance. Any necessary modifications will be conducted at SWFSC and tested in the SWFSC Ocean Technology Development Tank and/or locally at Mission Bay before tethering in the CINMS.

After consideration, we decided to only arm the mooring (Fig. 2) with one acoustic release, not two. This was in part due to the cost of adding a second acoustic release to each mooring, and in part because we learned that in the locations and at the depths we were working in, the risk of equipment loss due to lack of redundancy was minimal. Additionally, we developed a simple and low-cost 'weight retrieval device' that was tested offshore San Diego and implemented at our Santa Rosa Island site (Fig. 3). This site provides a SCUBA diveable depth, for manual assistance in case of entanglement, or release failure. The weight retrieval device performed well and variants of the design continue to be used on additional deployments in CINMS and in other west coast sanctuaries. Lastly, our initial mooring design had corrosion issues due to dissimilar metals, and future deployments minimized the use of metals.

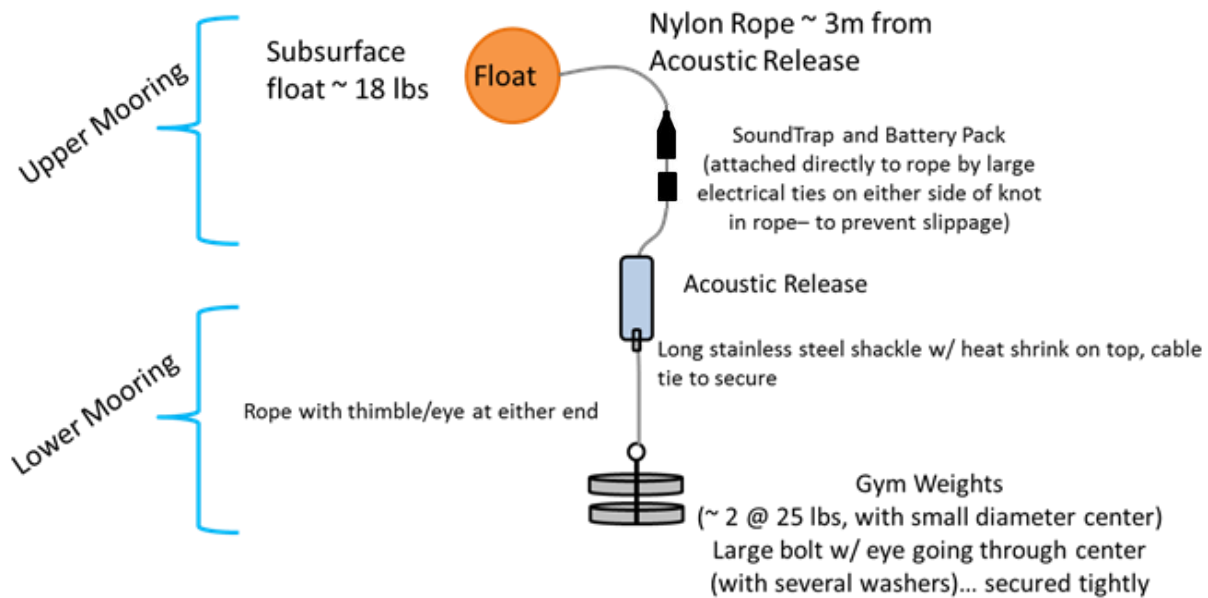


Figure 2. CINMS-SWFSC seafloor hydrophone mooring design.

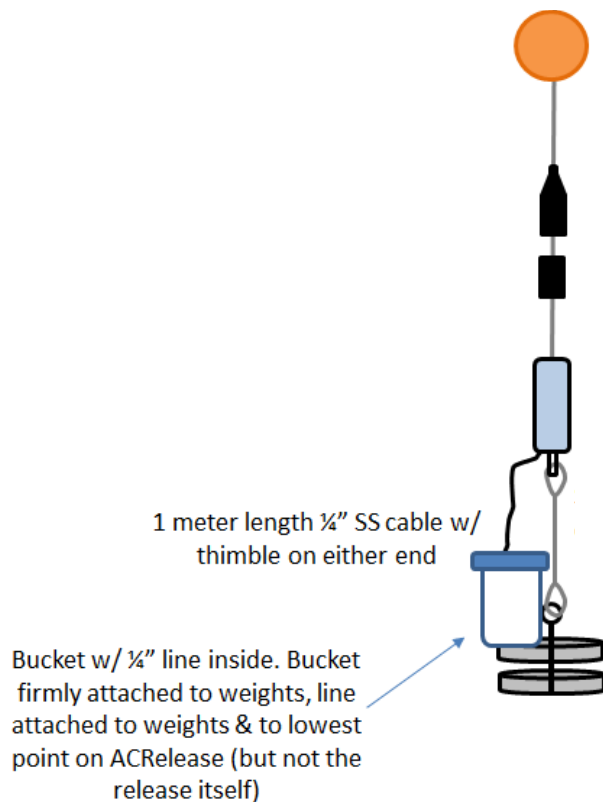


Figure 3. CINMS-SWFSC simple, low-cost weight retrieval bucket system.

1.4 Field survey (SWFSC & CINMS)

The field survey will consist of deployment and retrieval of the recording equipment in the two proposed locations. Field operations will be conducted aboard the CINMS' R/V *Shearwater*. We have estimated two days for deployment, and two days for recovery. This includes mobilization, transit, and operations. Timing of the start/end of field survey/data collection depends on completion of 1.1-1.3, availability of the R/V and researchers, and weather. Ideally the deployment will begin in April, before squid fishing opens in the Southern California Bight. CINMS will support additional maintenance of this equipment if and when resources allow.

Table 1. Field activities log.

Deployment	Retrieval
May 30, 2018	Aug. 9, 2018
Aug. 9, 2018	Oct. 31, 2018

The two sampling locations were as proposed: west of San Miguel Island (34.0856, -120.5232) at ~75m; north of Santa Rosa Island (34.0438, -120.0811) at ~20m. We originally budgeted for four days at sea (DAS), however we completed the project using three (Table 1). Therefore, we donated the remaining DAS on R/V *Shearwater* to the NOAA Noise Reference Station (NRS) project to re-deploy the NRS05 listening station south of Santa Cruz Island (33.8979, -119.5450) at ~1000m on Jan. 17, 2018.

Data Analysis

2.1 Develop seal bomb auto-detector (SWFSC)

Sound sources may have a physical, biological, or anthropogenic origin. In order to discriminate sounds SWFSC personnel are developing a statistical package (in R) called "BANTER" for acoustic classification (Rankin et al. 2016). This robust hierarchical random forest approach to automatically categorizing sound can generally be applied to any unique sound, including anthropogenic noise such as that produced by seal bombs. This approach will allow us to record the entire soundscape, but differentiate seal bombs from other noise inputs, allowing for targeted comparisons across sites. The advantages of BANTER are that it requires little data manipulation on the front-end, relying on output from a suite of generic automated pulse and tonal detectors within PAMguard software (Gillespie et al. 2008). SWFSC will first manually scan a section of data to identify seal bombs. Then, a relevant subsample of the entire dataset will be analyzed with a suite of generic automated pulse and tonal detectors within PAMguard. A seal-bomb classifier will be developed using the output of this suite of detectors, using the BANTER approach.

See next section for 2.1 results.

2.2 Identify the best approach for identifying seal bombs (SWFSC)

SWFSC will compare the correct classification of seal bombs using the BANTER approach vs. the independent automated detections within PAMguard (pulse and tonal detectors) to determine if the

additional effort required by BANTER improves classification. Likewise, these will be compared with the manual detection to determine the best approach for this and future studies.

See next section for 2.2 results.

2.3 Quantify seal bomb noise (SWFSC)

SWFSC will filter an appropriate subsample of data (~6 weeks) from the two SoundTrap listening stations to identify and quantify the temporal/spectral characteristics of seal bomb noise across stations. The time frame will allow us to examine seal bomb noise before and after the initiation of squid fishing season in one heavily fished area (north of Santa Rosa Island) and one minimally fished area (north west of San Miguel Island). We will specify methods for analyzing NRS05 data to quantify temporal variation in seal bomb noise; if resources allow, we will initiate this analysis with PMEL partners.

See next section for 2.3 results. Note that we omitted the NRS05 analysis task because that listening station samples at 1000m on the south side of Santa Cruz Island, which is a much different sampling regime than this project's shallow water sampling effort. Further, the data from that station for the Nov. 2014-Nov. 2017 time period was lost. The NRS05 data from Jan 2018-Dec 2019 was only just recovered, and so falls outside of the scope of this project. The NOAA-Navy *SanctSound* program (<https://sanctuaries.noaa.gov/science/monitoring/sound/>) does intend to integrate NRS data with auxiliary sound monitoring data, such as from this project, and specifically for sound sources like noise from fisheries.

2.4 (CINMS)

CINMS will work with the Office of Protected Resources (point person: Kristy Long) and, as appropriate, the NOAA Marine Mammal Non-Lethal Deterrents working group to share this study's findings. NOAA acoustic deterrent management tools will be used to identify potential mortality/serious injury of marine mammals based on the cumulative effect in regions of high fishing effort. This study will provide updated duty cycle data to reflect cumulative effects; source levels will consider estimates from previous studies (e.g., SIO). CINMS has been monitoring marine mammal species distributions, particularly marine mammal interactions with the shipping industry, for almost twenty years (Polefka et al. 2004; Redfern et al. 2017). Integrating available data sets in a systematic way will enable better examination of potential seal bomb impacts to marine mammals that are relevant to ear injury thresholds NOAA has identified. In addition, it will aid NOAA's continued research on quantifying and managing for cumulative impacts to marine life.

Lindsey met with Kristy in Aug of 2017, prior to the official start of this project. Kristy explained that the NOAA Marine Mammal Non-Lethal Deterrents working group was using seal bomb data products from SIO, and welcomes results from new projects in new areas. This project, and a follow-on project in Monterey Bay National Marine Sanctuary (MBNMS) discussed later, offer those types of sought-after information. In Feb 2019 Anne Simonis briefly NMFS acoustic personnel on this project. Additionally, Shannon Rankin and Lindsey Peavey Reeves participated in the NOAA Acoustic Coordination meeting in April 2019 where the working group's activities and what NMFS is currently doing to address seal bomb use was discussed. Once more results become available from the MBNMS and *SanctSound* projects, it will be combined with results from this project and offered to the working group.

2.6 Archive data (wav and metadata) to InPort (SWFSC)

Data will be archived to InPort following PARR recommendations, pending approval for release by the U.S. Navy.

NOAA is developing passive acoustic archive capacity at NCEI; however, they are not yet prepared to receive NOAA data beyond pilot intakes. This project's data has been released by the Navy after their review and is currently stored at SWFSC and can be made available to users as needed. NCEI archiving will be completed when the capacity and/or funds become available.

2.7 Prepare report and identify future monitoring opportunities (CINMS & SWFSC)

The final report will present results from each of the tasks and explore opportunities for continued monitoring. The report will be published as a NOAA Technical Memorandum and in a peer-reviewed scientific journal(s), if applicable.

Content from this report will appear in a *Marine Sanctuaries Conservation Series* technical report authored by this project's PIs and others describing various soundscape monitoring activities in CINMS. Apparent from the results presented below, there is more work to be done across west coast sanctuaries to converge on a reliable seal bomb detector, and to better anticipate and track seal bomb sources, such as squid fishing. Towards this objective, the SWFSC PIs on this project have forged a follow-on collaborative project with similar objectives, also funded by OAP, with MBNMS.

Results

2.1

For the two deployments (05/30/2018 - 10/31/2018), an analyst (Anne Simonis) manually identified 170 explosions in the San Miguel recordings, and 3 explosions in the Santa Rosa recordings (Fig. 4).

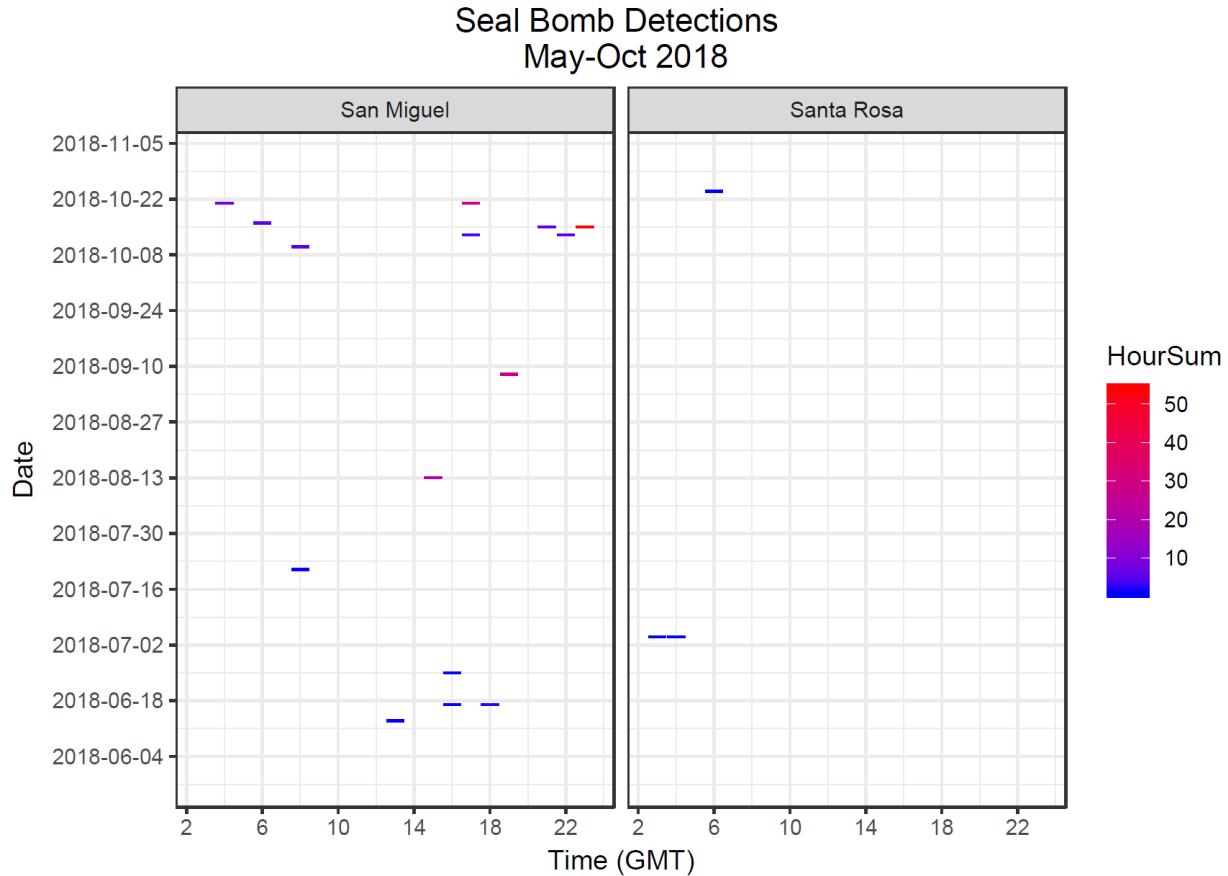


Figure 4. Hourly counts of verified seal bomb explosions from San Miguel and Santa Rosa SoundTraps from June - October 31, 2018. Colors indicate the total number of explosions per hour.

A subset of 99 explosion detections was extracted from the San Miguel recordings from 10/15/2018 and 10/21/2018. The built-in PAMguard pulse and tonal detectors were optimized for the observed explosions, however the false positive and false negative detection rates in each were unacceptably high. Instead, the PAMguard spectrogram correlation and energy sum detection methods were explored.

Table 2. Optimized Energy Sum Detector Parameters (PamGuard version 2.00.16e)

Lower Frequency Bound (Hz)	300
Upper Frequency Bound (Hz)	700
Threshold	0.01
Min time over threshold (s)	0.005
Max time over threshold (s)	NA
Min IDI (s)	0.3

Table 3. Optimized Spectrogram Correlation Detector Parameters (PamGuard version 2.00.16e)

t0 (s)	0
f0 (Hz)	1000
t1 (s)	0.02
f1 (Hz)	700
Kernal Width (Hz)	100
Threshold	0.03
Min time over threshold	2e-6
Min time before next detection	0.3

Detector parameters were optimized based on the 99 manually identified detections, and tested on a subset of data from 10/15/2018 - 10/21/2018. The true positive rates were 88% and 76% for the energy sum and spectrogram correlation detectors, respectively. The false positive rates were 96% and 98% for the energy sum and spectrogram correlation detectors, respectively. Sources of false detections included sea lion vocalization, fish grunts, debris hitting the hydrophone, and impulses at the onset of some audio files. The optimized detector parameters are shown in Tables 2 & 3.

Automatic detections were classified as “True” or “False” explosion detections using the SoundSort software (<https://github.com/macster110/aipam>). SoundSort is a Java and JavaFX based program which allows users to visualize, listen to, and cluster sound clips to support annotation.

A suite of spectral and temporal features were calculated for each detection, and the annotated energy sum and spectrogram correlation detections were used to train and test a BANTER model. A summary of the BANTER model is shown in the confusion matrix in Table 4.

Table 4. BANTER Confusion Matrix. Rows indicate actual class, columns indicate predicted class.

	False	True	Percent Correct
False	1944	1162	62.6 %
True	17	65	79.3%
Overall			63.0

2.2

The false positive rate in the BANTER model is currently too high to be useful, and suggests that there are still improvements that can be made to the model. Increasing the sample size of the ground truth detections will likely go far to enhance BANTER performance. Additional thresholds or classes for explosion types (e.g. “gunshot”, “seal bomb”) may also be useful additions to the model. Researchers at Monterey Bay Aquarium Research Institute (MBARI) have recently developed an automated spectrogram correlation detector based on multiple kernels, which may be helpful to reduce false detections before the BANTER classification step. Currently, manual methods are still the best approach to monitor long term acoustic recordings for the presence of seal bomb explosions.

2.3

There was some variation in the spectral and temporal characteristics of explosions (Fig. 5), although no clear temporal pattern in different signal types was evident throughout the recording period. This variation may be due to the propagation effects of explosions detected at varying ranges, or it may also indicate that there are multiple types of explosions present near the recording locations.

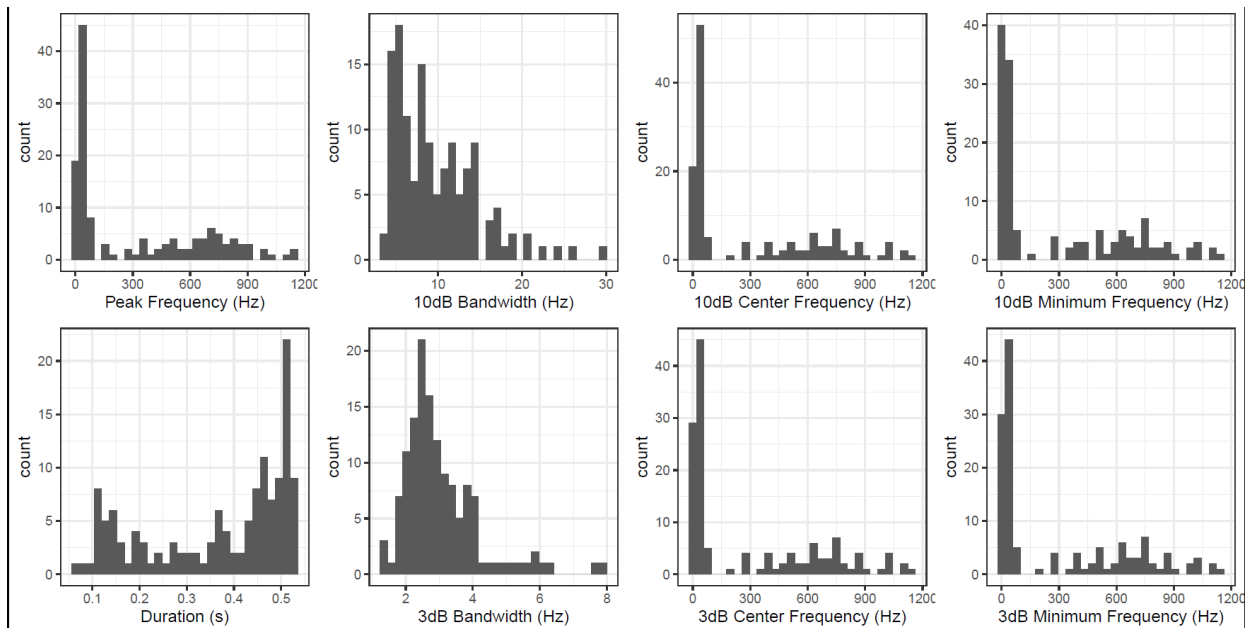


Figure 5. Histograms of spectral and temporal characteristics of verified seal bomb explosions from San Miguel Island sampling location.

Market squid fishing around the Channel Islands has been greatly reduced since 2014, as the target species shifted north in response to warm water events that began in 2013. For comparison, in 2010 the Ventura region brought in 126 million pounds of squid valued at \$33.7 million, and decreased to 34 million pounds valued at \$16.8 million in 2016. It appears that squid have not yet returned to the region—by 2018 landings were less than half of 2010 landings. It's difficult to track squid fishing vessels because they are a pelagic fishery that is not required to use NOAA Vessel Monitoring Systems (VMS). These vessels are typically smaller than the required length for Automatic Identification System (AIS) tracking technology. When we realized how difficult it is to use routine vessel tracking approaches for this objective, we began exploring the utility of detecting squid fishing vessel lights in nighttime satellite data. This approach has been used in other areas such as Chile, however we have not converged on a successful algorithm for our study region, and this project component is still exploratory.

In contrast, squid are easier to track given they are voracious predators that follow cool, nutrient-rich waters where they can find prey. We understand squid have shifted north to find cooler water with abundant prey. In our study region, our objective is to overlay squid fishing effort with seal bomb detections to see if we are able to build the hypothesis that squid fishing vessels are the source of seal bomb detections, and how many. It is not surprising that since squid fishing effort was so low in the study region during this study time period, so were seal bomb detections. This situation does support our hypothesis, however with such a low sample size of seal bomb detections, it makes our goals of developing a seal bomb detector and refining the correlation of squid fishing and detections very difficult.

To continue this study's objectives, we have moved the project north to MBNMS, where MBARI has had good luck detecting seal bombs on their cabled hydrophone, Monterey Accelerated Research System (MARS). "Investigation into the effect of seal bombs on harbor porpoise in Monterey Bay" is a collaboration also funded by OAP, and the collaborating organizations include SWFSC, MBNMS, and Naval Postgraduate School (NPS). By design, both the CINMS and the MBNMS seal bomb studies directly overlap and integrate with the ongoing *SanctSound* ONMS-Navy soundscape monitoring program. Namely, 1) the two sampling locations in this study have continued as *SanctSound* listening stations for continuity. 2) The MBNMS sampling locations were selected to compliment the *SanctSound* listening stations in MBNMS, and the NPS partnership builds on their role in leading the *SanctSound* efforts in MBNMS. The SoundTrap300s from this CINMS project are currently deployed for the MBNMS survey to allow for an increase in the sampling effort and therefore an increase in the potential validity of results across both projects.

Education and Outreach

Although education activities were not proposed, both NOAA sanctuaries and science centers are committed to outreach. Below is a bulleted list of some of our activities. Please be in touch with any questions or if you'd like more details regarding any of the following.

- NOAA Teacher-At-Sea (Jenn Sportsman, Ernest Righetti High School) and students participation in fieldwork on *R/V Shearwater* (multiple instances)
- CINMS and WCR sanctuaries Education Team meetings; collaborative education project proposal submissions (none were funded): July 2018, Aug 2018, March 2019 (among others)

- Scientist (Anne Simonis) classroom visit at Ernest Righetti High School
- Curricular materials provided to Jenn Sportsman
 - lesson plans utilizing authentic acoustic data
 - current marine science & conservation news clips
 - scientific literature on topics of interest or authored by Latino authors
- Organize campus visit and tours of Scripps Institution of Oceanography & Whale Acoustics Lab for Jenn Sportsman and 40 students
- Example CINMS soundscape developed and available online: <https://nature-other.ambient-mixer.com/channel-islands-nms-soundscape>
- NOAA ONMS webinar, “Listening to “See” Beneath the Waves: Soundscape monitoring in Channel Islands National Marine Sanctuary”, presented by Lindsey Peavey Reeves in April 2019. Archive available: <https://sanctuaries.noaa.gov/education/teachers/webinar-series-archives.html>
- Poster presentation at OceanObs’19 in Sept. 2019 (PDF of poster submitted as supporting material for this report)